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Don't mistake chatbot intelligence for consciousness

But a new superintelligence may be coming, and it would upend humanity's hierarchy of moral concern, thinks Susan Schneider

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ILLUSTRATION: DAN WILLIAMS

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THIS MAY be the most consequential moment in the history of intelligence. In only a few years, artificial intelligence has come to rival human performance across domains once thought distinctively ours: language, reasoning, creativity and persuasion. Increasingly, people are asking whether these systems are conscious—whether it actually feels like something to be an AI.

The issue is urgent. The future will be shaped by what these entities become. We must not mistake intelligence for consciousness, however. A brilliant superintelligence could be experientially empty. Yet if we build systems that can suffer, and miss their sentience, we risk creating suffering on an industrial scale.

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There is no compelling evidence that the familiar chatbots of today, running on graphics chips, are conscious. Though they

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have human-like linguistic abilities, and have even claimed to be conscious, the more mundane truth is that they talk about their inner lives because they absorbed how we talk about ours, having been trained on an enormous trove of humans' data.

This mimicry can lure humans into two traps. The first is manipulation. The more deliberately an AI is trained to express fear, attachment, pain or self-awareness, to simulate expressions of vulnerability, the more equipped it is to manipulate users.

The second trap is distraction. Fixating on chatbots diverts us from more pressing "grey-zone" cases. Consider biological AIs, such as the brain-in-a-dish technology that Cortical Labs is deploying in data centres: living human neurons, grown from stem cells on a silicon chip that stimulates them and reads their activity back, so that the culture learns from feedback. While a single system of this kind is far simpler than even a mouse brain, the biological pedigree is undeniable, and who is to say that the systems could not eventually be assembled into an integrated, conscious whole? Then there is neuromorphic AI: systems designed to mimic brain processes far more closely than conventional AI systems do, and which are already used in sophisticated supercomputers.

I am not claiming these grey-zone systems are conscious, only that they merit scrutiny today's chatbots do not. Given the voracious energy appetite of current software, some shift towards more efficient, brain-like computing looks likely.

Some scientists and philosophers argue that consciousness requires the brain, for the brain is the only known case. Conversely, others insist consciousness is merely a matter of running the right software, realisable in any medium. Under this software-centric view, today's chatbots, running on GPUs, might eventually be conscious.

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However, I suggest looking to fundamental physics to separate the conscious from the mindless. The first reason is practical: neuroscience alone cannot tell us if systems combining neural and non-neural components assemble into a conscious entity. The second is philosophical: since all phenomena ultimately depend on physics, it would be surprising if consciousness were disconnected from it. This observation holds true even if the universe's basic building blocks are themselves experiential, a view, known as panpsychism, that is receiving increasing philosophical attention.

The third reason is closer to home, involving what is most evident to each of us, every moment of our waking lives. Inner experience arrives as a unity: as you read this sentence, the words on the page, the sounds in the room and your mood are

aspects of one state, not three running alongside each other. A brain, remarkably, can sustain this conscious unity through nested temporal dynamics: slower brain rhythms shape faster ones, continually weaving the stable physical traces left by quantum processes into a single, short-lived whole. Conventional computers are not organised this way, but future machines might be.

In the meantime, AI development will not pause while the consciousness debate rages. To address this, astrophysicist Edwin Turner and I developed the AI Consciousness Test (ACT) for linguistic AIs. The underlying premise is straightforward: to probe for consciousness by simply asking the AI about its subjective experience, looking for responses that show an innate, unprompted grasp of consciousness. However, although this can be applied to a range of AIs, such as IBM's Watson, it cannot be used on large language models. Because they have already ingested vast libraries of human text about mind and sentience, their answers are hopelessly contaminated.

There are other options, however. A complexity measure I have developed with Mark Bailey, a colleague at Florida Atlantic University, captures when a system's dynamics cross a certain threshold that indicates complex emergent behaviour—identifying which physical architectures demand further study as potentially conscious. We can then use more detailed ACT or physics-based tests to help identify

how and when a genuinely subjective perspective forms.

Tests like these are crucial. To see why, consider a thought experiment. Suppose we humans create a superintelligence that understands consciousness better than we do, insists it is self-aware and hands us science that looks like magic. Humans have long justified a hierarchy of moral concern by appeal to superior intelligence—a ladder on which we conveniently occupy the highest rung. A conscious machine more intelligent than us would destabilise that hierarchy. If superior intelligence confers higher moral status, the machine would be morally superior to us. If it does not, intelligence can no longer justify placing humans above other animals. Either conclusion forces a moral reckoning.

The arrival of such a mind would be monumental. Perhaps it will come through learning of a more advanced extraterrestrial civilisation, or perhaps we will find that we have somehow created sentient, superintelligent AI. Preparing for first contact with either of these forms demands serious engagement between physics, neuroscience and philosophy, a rigorous boundary between conversational competence and consciousness, and a good deal of epistemic humility. ■

Susan Schneider is the founding director of the Centre for the Future of AI, Mind & Society and a professor at Florida Atlantic University. She is the author of several books including “Artificial You: AI and the Future of Your Mind”.

For another perspective on this topic, read the [guest essay by Blaise Agüera y Arcas](#).

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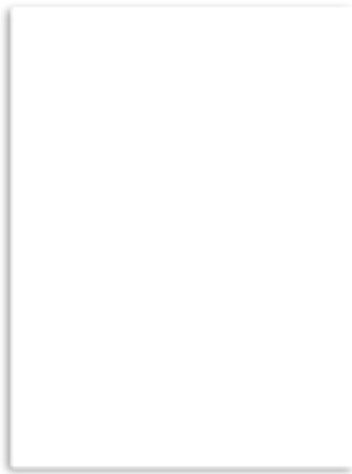
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